

Oween Barranzuela Carrasco

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Profile

AI B.Sc. student with hands-on Machine Learning experience in multivariate time-series and sensor data, currently conducting a Bachelor's thesis at BSH Hausgeräte GmbH on ML-based analysis of spectrometer data. Experienced in data preprocessing, classification, model evaluation, and robustness analysis in industrial and research environments. Interested in anomaly detection, sensor data quality, and robust Machine Learning for real-world AI systems.

Education

- **Deggendorf Institute of Technology** *10/2023 - Present*
  Artificial Intelligence, B. Sc. Deggendorf, Germany
  - GPA: 2.2 (7th Semester); focus on Machine Learning, Deep Learning, Software Engineering and Python Programming.

- **Vilnius University of Applied Sciences** *09/2025 – 11/2025*
  Erasmus+ Blended Intensive Programme — Student Management System Vilnius, Lithuania
  - Selected for an Erasmus+ programme to design and prototype a student management system, focusing on database architecture, authentication modules, and academic information workflows using Oracle APEX.

- **Selçuk University** *2019–2023*
  International Business and Relations, B. A. Konya, Turkey
  - GPA: 3.64/4.00

Relevant Experience

- **BSH Hausgeräte GmbH** *09/2026 – Present*
  Bachelor's Thesis – ML Models for Soil Detection via Spectrometer Data Dillingen a.d. Donau, Germany
  - Own the end-to-end ML pipeline for detecting soil on dishwasher surfaces from spectrometer sensor data, from experiment design through model selection and evaluation.
  - Plan and execute data collection scenarios in-machine dishwasher conditions to build a representative dataset.
  - Train and benchmark multiple ML models using cross-validation, assessing robustness and generalization across the two environments.
  - Document methodology and results in an academic thesis, covering experimental design, model comparison, and findings.

- **BSH Hausgeräte GmbH** *04/2026 – 09/2026*
  AI Intern – Pre-Development for Dishwashers Dillingen a.d. Donau, Germany
  - Built and evaluated time-series classification models (ROCKET/MiniRocket) on multivariate sensor data, benchmarking against baseline architectures to improve classification accuracy.
  - Designed data augmentation pipelines to address gaps in temperature and conductivity sensor data, improving test accuracy by 2% and increasing model robustness on limited industrial datasets.
  - Documented experiment workflows and model evaluation metrics, enabling reproducible validation across iterations within the AI Engineering team.

- **Deggendorf Institute of Technology** *10/2025 – 03/2026*
  Student Research Assistant – Machine Learning for Sensor Data Deggendorf, Germany
  - Cleaned and preprocessed 1+ year of multi-sensor water-quality time-series data using Python, pandas, NumPy and scikit-learn.
  - Built turbidity and nitrate prediction pipelines, fusing sensor streams with meteorological data and engineering lag, rolling-window and cyclical time features.
  - Benchmarked 10 regression models (Linear/Ridge/ElasticNet/Huber, Random Forest/ExtraTrees, Gradient Boosting) on a future holdout window using MAE, RMSE and R^2.
  - Achieved $R^2 = 0.52$ and **RMSE** ≈ 0.29 for turbidity on the most recent 20-day window via recency-weighted training.
  - Presented experiments, plots and model behavior in structured LaTeX reports to guide internal team decisions.

Academic Projects

- **ZeitZuHelfen – Voice-First Help Platform for Seniors** *2025*
  Tools: FastAPI, Telegram Bot API, Google Gemini 2.0 Flash, SQLite, SQLAlchemy, Pydantic
  - Developed a voice-first help request platform during HackaTUM 2025, connecting seniors in Munich with student helpers.
  - Implemented a FastAPI backend with user management, authentication and request CRUD operations using SQLite and SQLAlchemy.
  - Built a Telegram bot that processes voice messages and uses Google Gemini 2.0 Flash to extract structured help-request data.
  - Integrated Google Address Validation API and gTTS to validate German addresses and provide natural German voice responses.
- **Stone & Green-Area Detection – Embedded Object Detection** *2025*
  Tools: Python, OpenCV, YOLOv4, LabelImg, Raspberry Pi
  - Built an embedded computer-vision pipeline to detect stones and classify green areas for a small-scale outdoor navigation scenario.
  - Collected and manually annotated images in YOLO format, maintaining a clean dataset structure and train-validation split.
  - Trained and fine-tuned a YOLOv4 object detector, improving robustness with data augmentation techniques.
  - Deployed and tested the model on a Raspberry Pi setup, validating real-time inference behavior and reducing false positives.
- **IcebergAI – Full-Stack ML Web Application** *2025*
  Tools: Python, scikit-learn, React+Vite, Docker, GitLab
  - Built an end-to-end ML web application to predict Titanic passenger survival based on demographic and ticket features.
  - Implemented and evaluated classification models in Python with scikit-learn and exposed the prediction logic through a backend API.
  - Designed a responsive React+Vite frontend to collect user input and display prediction results clearly.
  - Containerized the application with Docker and coordinated a 7-member team using GitLab issues and merge requests.

Technical Skills

- **Programming & Data:** Python, SQL, pandas, NumPy
- **ML & AI:** scikit-learn, sktime, PyTorch, LangChain, time-series modeling, model evaluation (MSE, MAE, R^2)
- **Web & APIs:** FastAPI, Flask, Streamlit, React, Telegram Bot API, Oracle APEX
- **Databases & BI:** MySQL, PostgreSQL, SQLite, Power BI
- **Tools & DevOps:** Git, GitHub/GitLab, Docker, Linux, Jupyter, Google Colab, LaTeX, Grafana

Awards

- **HackaTUM 2025 – City of Munich Challenge** *11/2025*
  TUM Munich
  - Placed in the Top 10 teams with the *ZeitZuHelfen* project.
- **STIBET Scholarship** *2025*
  DAAD / THD
  - Awarded for outstanding commitment at Deggendorf Institute of Technology.
- **Erasmus+ Scholarship** *2025*
  Erasmus+
  - Funded participation in an Erasmus+ Blended Intensive Programme in Vilnius.

Training and Certifications

- LangChain – Develop AI Agents with LangChain and LangGraph (Udemy)
- Supervised Machine Learning: Regression and Classification (Coursera)

Languages

Spanish (Native), English (C1), German (C1), Turkish (C1), Portuguese (A1)